Εισαγωγή στη Ρομποτική ---- 4η Εργασία

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Task 1:

"Extract the kinematic and dynamics information from the URDF file"

There were many ways to extract information from the URDF file- via VS Code, with MATLAB or just in text. Utilizing all of them, we can extract these data:

Arm_link_0: It is the base of the robot with origin: (x y z) = (0 0 0.02725). This is also the CoM. There is no Rotational Inertia Matrix defined as the box is considered fixed.

Arm_link_1: Second "box" with origin: (x y z) = (0 0 0.0615). This is also the CoM. Its mass is:

$$m \approx 0.19$$

The Rotational Inertia Matrix is defined as:

$$I_b = I_i = \begin{matrix} 0.00027 & 0 & 0 \\ 0 & 0.00026 & 0 \\ 0 & 0 & 0.000065 \end{matrix}$$

Arm_link_2: Third "box" with origin: (x y z) = (0 0 0.1585). This is also the CoM. Its mass is:

$$m \approx 0.293$$

The Rotational Inertia Matrix is defined as:

$$I_b = I_i = \begin{matrix} 0.00025 & 0 & 0 \\ 0 & 0.00024 & 0 \\ 0 & 0 & 0.000092 \end{matrix}$$

Arm_link_3: Fourth "box" with origin: (x y z) = (0 0 0.101). This is also the CoM. Its mass is:

$$m \approx 0.22$$

The Rotational Inertia Matrix is defined as:

$$I_b = I_i = \begin{matrix} 0.00079 & 0 & 0 \\ 0 & 0.0007 & 0 \\ 0 & 0 & 0.000069 \end{matrix}$$

Arm_link_4: Fifth "box" with origin: (x y z) = (0 0 0.202). This is also the CoM. Its mass is:

$$m \approx 0.16$$

The Rotational Inertia Matrix is defined as:

$$I_b = I_i = \begin{matrix} 0.00037 & 0 & 0 \\ 0 & 0.00036 & 0 \\ 0 & 0 & 0.0000496 \end{matrix}$$

Arm_joint_1: It is the first joint of the robot; it connects the first "box" (base) with the second "box" of the robot. Its origin is at (x y z) = (0 0 0.0545), and it rotates only in axis z (0 0 1).

Arm_joint_2: It is the second joint of the robot; it connects the second "box" (base) with the third "box" of the robot. Its origin is at (x y z) = (0 0 0.123), and it rotates only in axis y (0 1 0).

Arm_joint_3: It is the third joint of the robot; it connects the third "box" (base) with the fourth "box" of the robot. Its origin is at (x y z) = (0 0 0.317), and it rotates only in axis y (0 1 0).

Arm_joint_4: It is the fourth joint of the robot; it connects the fourth "box" (base) with the fifth "box" of the robot. Its origin is at (x y z) = (0 0 0.202), and it rotates only in axis y (0 1 0).

Arm_joint_5: It is the fifth joint of the robot; it connects the fifth "box" (base) with the end-effector of the robot. Its origin is at (x y z) = (0 0 0.1605). It is a fixed joint which means that it does not rotate on any axis.

* By the term origin, we mean that we have a current point in space that is defined based on the previous one.

We can calculate the Screw axis of each joint:

$$S = \begin{pmatrix} r \\ \rho \end{pmatrix}$$

- r : rotation axis
- ρ = -r x q , q: translation vector

Thus, we have:

$$r1 = (0\ 0\ 1), \rho 1 = (0\ 0\ 0.0545) \rightarrow S_1 = (0\ 0\ 1\ 0\ 0\ 0)^T$$

$$r2 = (0\ 1\ 0), \rho 2 = \big(0\ 0\ (0.0545 + 0.123)\big) = (0\ 0\ 0.1775) \rightarrow S_2 = (0\ 1\ 0\ -0.1775\ 0\ 0)^T$$

$$r3 = (0\ 1\ 0), \rho 3 = (0\ 0\ 0.4945) \rightarrow S_3 = (0\ 1\ 0\ -0.4945\ 0\ 0)^T$$

$$r4 = (0\ 1\ 0), \rho 4 = (0\ 0\ 0.6965) \rightarrow S_4 = (0\ 1\ 0\ -0.6965\ 0\ 0)^T$$

The fifth joint has no Screw axis because it is a fixed joint.

Task 2:

"Create a dynamics simulator that is able to simulate the robotic manipulator as defined in the URDF"

Firstly, as the task asks for a simulation, we know that the equation we must compute at the end of the exercise is (forward dynamics):

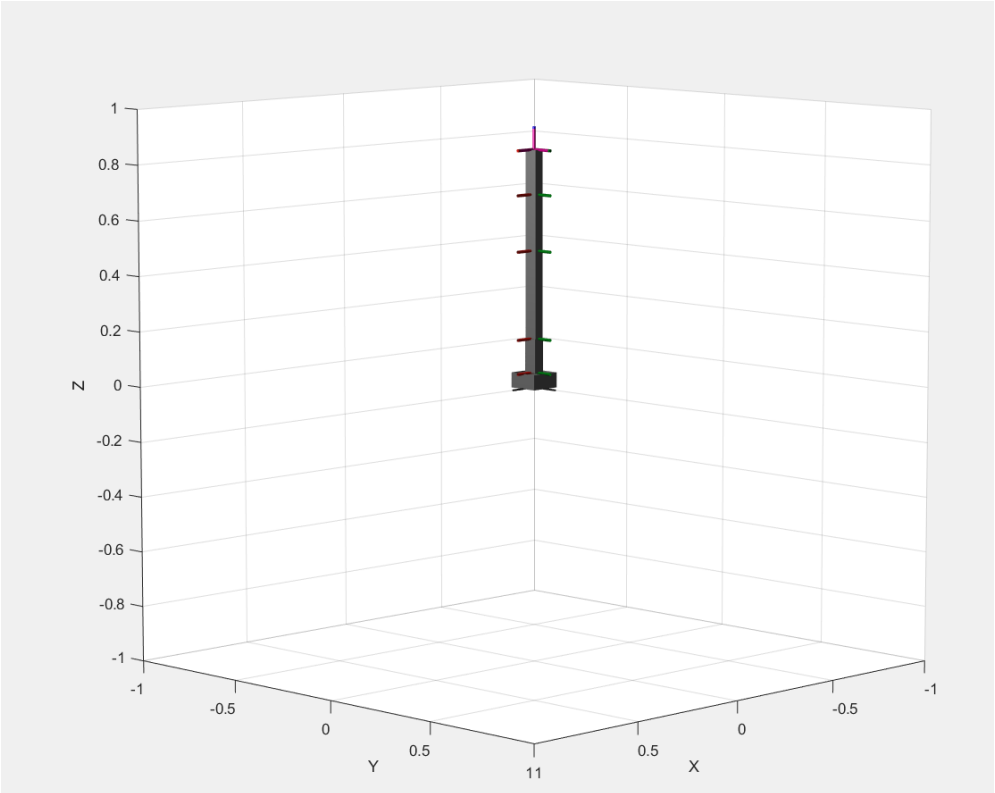
$$\ddot{q} = M(q)^{-1}(\tau - C(q, \dot{q}) - g(q))$$

The $Mass\ Matrix\ M(q), the\ Coriolis\ forces\ C(q, \dot{q})\ and\ the\ external\ force\ g(q)$ are called the "Dynamic quantities" of the dynamic simulation. To calculate them, we are going to use the Newton-Euler Inverse Dynamics algorithm (for the velocities and accelerations V , $\dot{V}$) and the closed-form formulation. The equations are in the code very analytical, so I will not explain further.

The whole idea is that we must denote the Screw axes Si in frame {i} (thus creating the Ai screw axes) , calculate all the Transformation Matrices (relative ones, with respect to the base, the inverse) and then apply the Newton-Euler method. We start with the forward pass to calculate the velocities and accelerations V , $\dot{V}$ and we finish with backward pass for the wrenches F (these are not needed). After that, we use the closed-form formulation to finally get the dynamic quantities that we want.

The joint positions and velocities $q, \dot{q}$ are updated using Euler integration because of its simplicity. I have given an initial non-zero value for the joint velocities of the robot for the integration to begin.

Start of the simulation:



End of the simulation:

